Fiscal Year:	FY 2007	**Task Last Updated:**	FY 10/03/2007
PI Name:	Caiozzo, Vincent J. Ph.D.		
Project Title:	Hypergravity Resistance Training: Countermeasure to Microgravity		
Division Name:	Human Research		
Program/Discipline:	NSBRI Teams		
Program/Discipline--Element/Subdiscipline:	NSBRI Teams--Muscle Alterations and Atrophy Team		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	92697-0001	**Congressional District:**	48
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2003 Biomedical Research & Countermeasures 03-OBPR-04
Start Date:	04/01/2004	**End Date:**	04/30/2008
No. of Post Docs:	2	**No. of PhD Degrees:**	0
No. of PhD Candidates:	1	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NSBRI
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	NOTE: Received NCE to 4/30/2008 per K. Major/NSBRI (3/08)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Baldwin, Kenneth (University of California, Irvine) Adams, Gregory (University of California, Irvine) Cavanagh, Peter (The Cleveland Clinic Foundation) Kreitenberg, Art (University of California, Irvine)		
Grant/Contract No.:	NCC 9-58-MA00403		
Performance Goal No.:			
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Task Description:	A program priority of NASA's Biomedical Research and Countermeasures Program (NRA-03-OBPR-04) is to determine the potential usefulness of artificial gravity as a countermeasure, especially with respect to skeletal muscle atrophy and loss of muscle function. The current proposal is a "proof-of-principle" of a unique countermeasure technology referred to as the "Space Cycle." The Space Cycle is a human powered centrifuge that can be used to generate various levels of artificial/hypergravity. The novelty of artificial gravity/hypergravity resistance training is that each element of the body is loaded proportional to the local gravitational field, and under hypergravity conditions muscles like those of the leg can be made to work against very high loads (e.g., + 2 body weights) without the need for external weights. The primary objective of this proposal is to use the Space Cycle to address the following general hypothesis: Artificial gravity can be used as a unique resistance training modality that acts as an effective countermeasure, preventing the loss of muscle mass and function that occurs due microgravity. In addressing this issue, a logical sequence of experiments is proposed with the following objectives: i) determine if squats performed under hypergravity conditions and without external weights can produce foot forces similar to those seen when performing squat resistance training (SRT) under normal 1 G conditions; ii) determine if squats performed under hypergravity conditions produce muscle adaptations similar to those seen using a squat resistance training program under normal 1 G conditions; and iii) determine if squat hypergravity resistance training (SHRT) program is an effective countermeasure to simulated microgravity. During the past two years, two studies were performed. The first study represents the initial step in testing the hypothesis that hypergravity can be used as a unique modality for resistance training, maintaining the health and function of skeletal muscle in microgravity. The primary objectives of this study were: i) to determine if subjects could perform squats under hypergravity conditions without developing motion sickness or illusory motion; ii) to measure foot forces while performing squats under hypergravity conditions; and iii) to determine the power required by the cyclist to produce 1.5, 2.0, 2.5, and 3.0 Gz. All studies were performed using a unique human-powered short-arm centrifuge, the Space Cycle. One of the centrifuge arms of the Space Cycle had a gondola or cage like configuration that allowed subjects to perform squats. Subjects (male n = 20; and female n = 16) performed squats under hypergravity conditions (1.5, 2.0, 2.5, and 3.0 Gz), and during these tests foot forces were monitored using the Pedar-X system. Foot forces measured under the hypergravity conditions were normalized to foot forces measured at 1 Gz, and these were referred to as relative foot forces (RFF). All of the subjects were able to perform the hypergravity squats without developing motion sickness, and illusory motion was minimized by having the subjects fix their sight on the leading edge of the base plate. All of the male and female subjects were able to perform squats at 1.5, 2.0, and 2.5 Gz. However, two male and two female subjects were not able to perform squats at 3.0 Gz because the loading at this hypergravity condition exceeded their strength. The mean relative foot forces at the highest Gz (i.e., 3.0) were 2.3 and 2.5 times greater than body weight for the male and female subjects, respectively. The work rate required to power the Space Cycle was a linear function of Gz, and is well within the aerobic scope of most untrained individuals. In conclusion, the findings of this study demonstrate that hypergravity can be used as an effective modality for loading skeletal muscle and that subjects can perform squat resistance exercise without becoming motion sick or experiencing illusory motion. These findings were presented at the 2006 Experimental Biology Meeting, the 2006 Annual Meeting of the American College of Sports Medicine, and the Investigator Workshop. Additionally, these data were published in Aviation, Space and Environmental Medicine 78:2-8, 2007. The objectives of the second study were to: 1) determine if hypergravity squats can produce foot forces similar to those measured while performing 10-repetition maximum (10-RM) squats under normal 1 Gz condition, and 2) compare the integrated EMG (iEMG) and goniometry of selected muscles and joints, respectively between hypergravity and 10-RM squats of similar foot forces. Subjects completed (male = 8 subjects; females = 5 subjects) ten 10-RM squats (97 ± 12 kg) and 10 hypergravity squats (3.2 ± 0.3 Gz; range of 3-3.75 Gz) under similar average total foot force conditions (104% 10-RM) in the same visit. Hypergravity squats were performed on a human powered short-arm centrifuge. Foot forces were monitored using insole force sensors. Hip and knee flexion/extension, plantar/dorsi flexion, and EMG of the erector spinae, bicep femoris, rectus femoris and gastrocnemius were also monitored during the squats. There were no differences in average total foot forces, average duration per repetition, and iEMG and range of motion of the selected muscles and joints, respectively between hypergravity and 10-RM squats except a 60% greater iEMG of the gastrocnemius muscles during hypergravity squats ($P < 0.05$). These results suggest that hypergravity squats are comparable to squats performed at 1 Gz, and may represent an important countermeasure to microgravity. Collectively, we have performed a series of sequential studies that will culminate in two training studies this coming year. The goal of the project entering the fourth year is to compare resistance training on the Space Cycle with that of SRT at 1 Gz.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	There are potentially several unique aspects of the Space Cycle that may have a direct benefit to earth based activities. Currently, it is unclear whether hypergravity may represent a unique loading mode that might enhance training and/or rehabilitation. The unique aspect of hypergravity is that each element is weighted in accordance to its specific gravitational field. In essence, this is like hanging a set of bar bells on every atom/molecule in the body. This distribution of load may be quite different than that of placing weights on a given body structure whereby there is a concentrated loading on that particular structure. Additionally, hypergravity may represent a unique modality for spinal cord injured patients. It is well known that many of these individuals have a reduced orthostatic tolerance. Some have suggested that exposing such individuals to hypergravity may help in this regard. We hope to explore this avenue of investigation in the coming years.
Task Progress:	he primary tasks of the past year were to: 1. To determine if male subjects could approach or exceed foot forces on the Space Cycle that approximated or exceeded those during a 10 RM squat. 2. To determine if female subjects could approach or exceed foot forces on the Space Cycle that approximated or exceeded those during a 10 RM squat. 3. To begin examining the effects of resistance training on the Space Cycle.
Bibliography Type:	Description: (Last Updated: 12/17/2008)

Abstracts for Journals and Proceedings	Caiozzo VJ, Yang Y, Pierre M, Kaplan A, Adams G, Kreitenberg A, Takahashi C, Cavanagh P. "Hypergravity resistance training as a countermeasure to microgravity: evolution of the space cycle." American College of Sports Medicine Annual Meeting, Denver, Colo., May 31-June 3, 2006. Med Sci Sports Exer. 2006 Jun;37:S72. , Jun-2006
Abstracts for Journals and Proceedings	Pierre M, Yang Y, Kaplan A, Cavanagh PR, Adams GR, Kreitenberg A, Takahashi C, Caiozzo VJ. "Dynamic biomechanical model of foot forces during hypergravity resistance training." American College of Sports Medicine Annual Meeting, Denver, Colo., May 31-June 3, 2006. Med Sci Sports Exerc. 2006 Jun;38:S65. , Jun-2006
Articles in Peer-reviewed Journals	Yang Y, Kaplan A, Pierre M, Adams G, Cavanagh P, Takahashi C, Kreitenberg A, Hicks J, Keyak J, Caiozzo V. "Space cycle: a human-powered centrifuge that can be used for hypergravity resistance training." Aviat Space Environ Med. 2007 Jan;78(1):2-9. PMID: 17225475 , Jan-2007